

Data acquisition (January–December 2025)

Ancillary data
MCD12Q1, boundaries

MODIS MOD11A1
LST Day / Night

MODIS MCD19A2
MAIAC AOD 550 nm

Sentinel-5P TROPOMI
CH₄, HCHO, O₃, UVAI,
NO₂, CO, SO₂

DoE ground stations
16 CAMS: PM_{2.5}, PM₁₀, gases

PRE-PROCESSING (GOOGLE EARTH ENGINE)

Quality control
AOD > 0, ≥ 2 obs/month, median; LST QC, error ≤ 2 K; SO₂ ≥ -0.001 mol m⁻²

Water masking (MCD12Q1 class 17)

Temporal compositing
Daily national means · Monthly · Seasonal · Annual

Coverage check
periods < 50% flagged

UHI intensity
urban – rural LST (10 km)

SPATIO-TEMPORAL ANALYSIS

Spatial mapping
Monthly, seasonal and annual maps

Temporal trends
Daily series, 15-day median

Seasonal statistics
Four seasons, national means

Diurnal contrast
Monthly LST Day vs Night

STATISTICAL ANALYSIS
0.05° grid, land cells (n = 4,901)

Correlation
Pearson and Spearman; annual and seasonal

Linear regression
LST vs each pollutant

PCA
Dominant components

GWR
LST ~ AOD + CH₄ + HCHO + NO₂ + CO

GROUND-BASED COMPARISON

Station sampling
3 × 3 pixels, monthly composites

Satellite vs station
Station-month and station-mean correlation

Air quality standards
PM vs WHO guideline and national standard

Results and discussion
Pollution patterns · Seasonality · LST–pollutant relationships · Local hotspots · Ground comparison